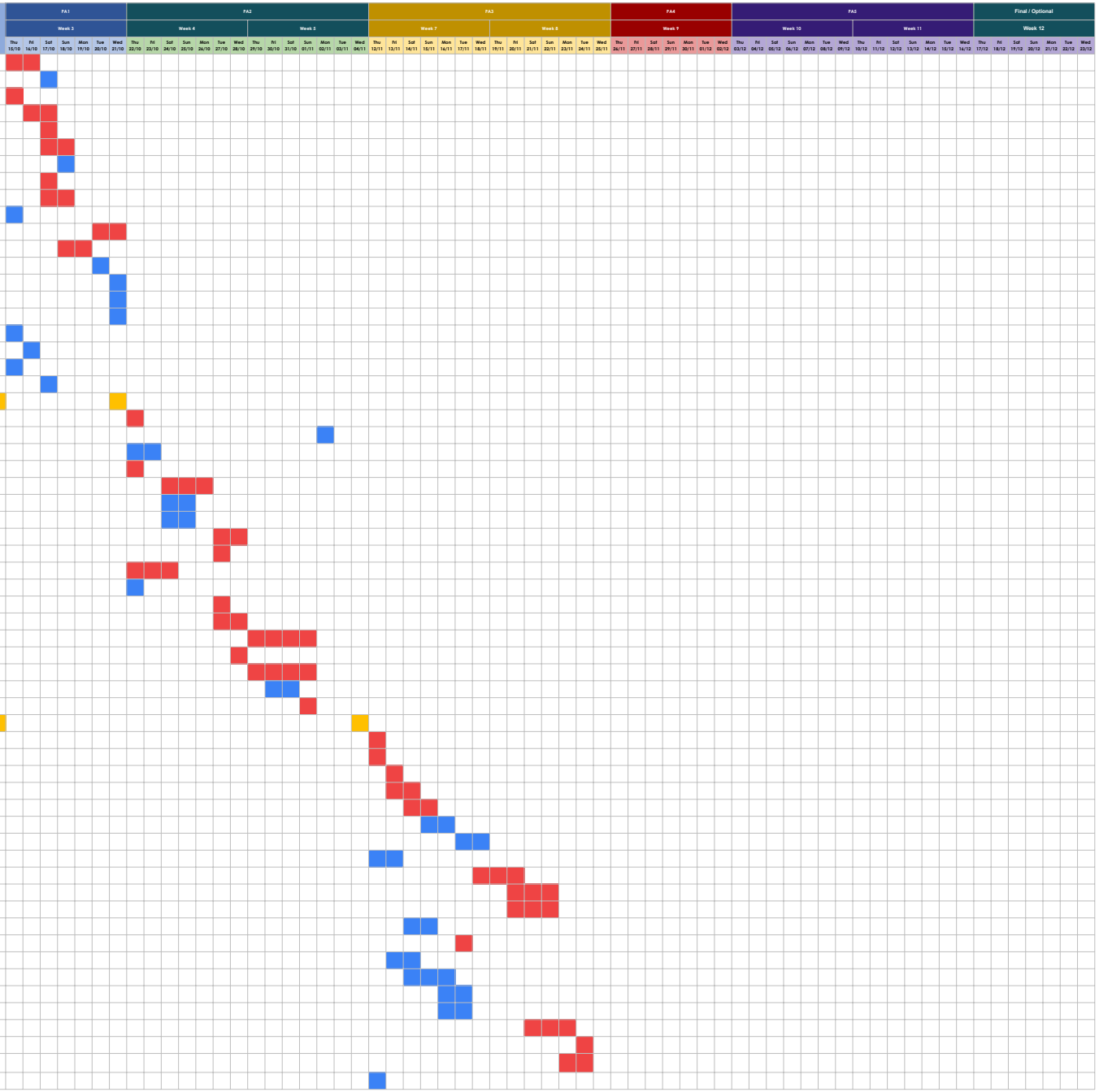
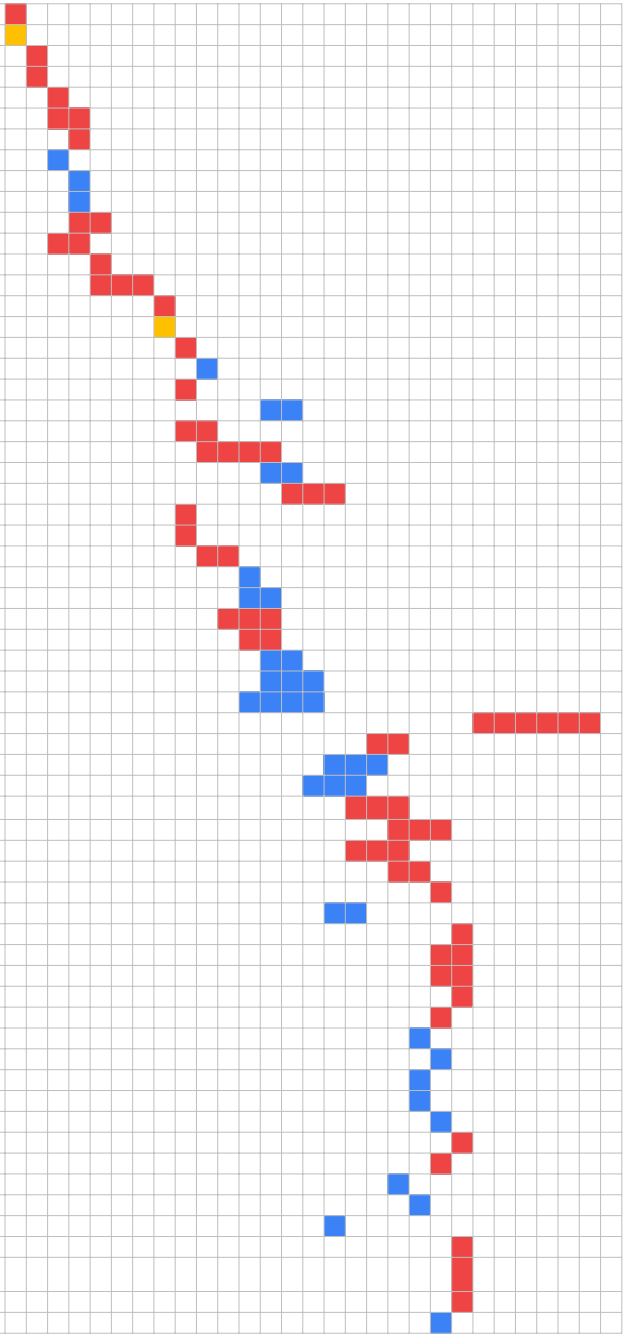
WBS ID	Task Title	Predecessors (WBS ID)	Owner	Start	Due	Duration (day)
1.1.1	Write the Preliminary Problem Statement		Lê Đoàn Nhật Huy	10/15/26	10/16/26	2
1.1.2	Write the Product Position Statement	1.1.1 FS, 1.1.4 FF	Lê Đoàn Nhật Huy	10/17/26	10/17/26	1
1.1.3	Document Stakeholders and Users		Lê Đoàn Nhật Huy	10/15/26	10/15/26	1
1.1.4	Define Scope and Key Features	1.1.1 SS+1d, 1.1.3 FS	Lê Đoàn Nhật Huy	10/16/26	10/17/26	2
1.2.1	Draw the Preliminary Architecture	1.1.4 FF	Trần Hoàng Minh Khang	10/17/26	10/17/26	1
1.2.2	Document the Technology Stack	1.2.1 SS	Trần Hoàng Minh Khang	10/17/26	10/18/26	2
1.2.3	Document Hardware and Operating Environment	1.2.2 FF	Trần Hoàng Minh Khang	10/18/26	10/18/26	1
1.3.1	Define Process and Phase Gates	1.1.4 FF	Huỳnh Trung Ngân	10/17/26	10/17/26	1
1.3.2	Create the Work Breakdown Structure	1.1.4 SS+1d, 1.3.1 SS	Huỳnh Trung Ngân	10/17/26	10/18/26	2
1.3.3	Document Roles and Responsibilities		Huỳnh Trung Ngân	10/15/26	10/15/26	1
1.3.4	Prepare the Schedule and Critical Path	1.3.5 FS, 1.4.3 FF	Trương Hiển Long	10/20/26	10/21/26	2
1.3.5	Prepare Work Package Estimation	1.3.2 SS+1d, 1.2.2 SS+1d	Trương Hiển Long	10/18/26	10/19/26	2
1.3.6	Prepare Human Resources and Cost Estimate	1.3.5 FS, 1.3.3 FS	Trương Hiển Long	10/20/26	10/20/26	1
1.4.1	Identify project risks	1.1.4 FS, 1.2.2 FS, 1.3.5 FS	Trần Hoàng Minh Khang	10/21/26	10/21/26	1
1.4.2	Calculate risk exposure	1.4.1 SS	Lê Đoàn Nhật Huy	10/21/26	10/21/26	1
1.4.3	Document risk responses and owners	1.4.2 FF	Lê Đoàn Nhật Huy	10/21/26	10/21/26	1
1.5.1	Set up communication channels		Huỳnh Trung Ngân	10/15/26	10/15/26	1
1.5.2	Set up the task board	1.5.1 FS, 1.3.3 FS	Huỳnh Trung Ngân	10/16/26	10/16/26	1
1.5.3	Set up the repository and collaboration rules		Trương Hiển Long	10/15/26	10/15/26	1
1.5.4	Document Tools and Collaboration Setup	1.5.2 FS, 1.5.3 FS	Trương Hiển Long	10/17/26	10/17/26	1
Milestone: PA1 Submission						
2.1.1	Write Purpose and Scope	1.1.2 FS, 1.3.4 FS, 1.4.3 FS	Huỳnh Trung Ngân	10/22/26	10/22/26	1
2.1.2	Prepare the Glossary	2.5.2 FS	Trương Hiển Long	11/2/26	11/2/26	1
2.1.3	Elaborate the Problem Statement	1.1.1 FS, 2.1.1 SS	Huỳnh Trung Ngân	10/22/26	10/23/26	2
2.1.4	Document Stakeholders	1.1.3 FS, 2.1.1 SS	Huỳnh Trung Ngân	10/22/26	10/22/26	1
2.2.1	Document Functional Requirements	2.2.6 SS+2d	Lê Đoàn Nhật Huy	10/24/26	10/26/26	3
2.2.2	Document Non-Functional Requirements	2.2.6 SS+2d	Trương Hiển Long	10/24/26	10/25/26	2
2.2.3	Document Domain Requirements and Business Rules	2.2.6 SS+2d	Lê Đoàn Nhật Huy	10/24/26	10/25/26	2
2.2.4	Define Acceptance Criteria	2.2.1 FS, 2.2.5 SS	Huỳnh Trung Ngân	10/27/26	10/28/26	2
2.2.5	Prioritise requirements and record sources	2.2.1 FS, 2.2.2 FS, 2.2.3 FS	Huỳnh Trung Ngân	10/27/26	10/27/26	1
2.2.6	Elicit and analyse requirements	2.2.1 SS, 2.2.4 SS	Lê Đoàn Nhật Huy	10/22/26	10/26/26	3
2.3.1	Identify Actors	2.2.4 SS	Trần Hoàng Minh Khang	10/22/26	10/22/26	1
2.3.2	Prepare the Use Case List	2.3.1 FS, 2.2.1 FS, 2.2.3 FS	Lê Đoàn Nhật Huy	10/27/26	10/27/26	1
2.3.3	Draw the Use Case Diagram	2.3.2 SS	Trần Hoàng Minh Khang	10/27/26	10/28/26	2
2.3.4	Write Use Case Specifications	2.3.3 FS, 2.2.4 FS	Trương Hiển Long	10/29/26	11/1/26	4
2.4.1	Prepare the Screen List	2.3.2 FS	Trần Hoàng Minh Khang	10/28/26	10/28/26	1
2.4.2	Create and check the Prototype and Mockups	2.4.1 FS, 2.3.3 FS	Trần Hoàng Minh Khang	10/29/26	11/1/26	4
2.5.1	Prepare the Requirements Traceability Matrix	2.2.5 FS, 2.3.2 FS, 2.4.2 FS	Huỳnh Trung Ngân	10/30/26	10/31/26	2
2.5.2	Conduct and document the Requirements Review	2.5.1 FS, 2.3.4 FF, 2.4.2 FF	Lê Đoàn Nhật Huy	11/1/26	11/1/26	1
Milestone: PA2 Submission						
3.1.1	Document Design Goals and Constraints	2.5.2 FS	Trần Hoàng Minh Khang	11/12/26	11/12/26	1
3.1.2	Document the Architectural Style	3.1.1 SS	Trần Hoàng Minh Khang	11/12/26	11/12/26	1
3.1.3	Update the System Context Diagram	3.1.1 FS	Trần Hoàng Minh Khang	11/13/26	11/13/26	1
3.1.4	Draw the Container Diagram	3.1.3 SS, 3.1.2 FS	Trần Hoàng Minh Khang	11/13/26	11/14/26	2
3.1.5	Draw the Component Diagram	3.1.4 SS+1d	Trần Hoàng Minh Khang	11/14/26	11/15/26	2
3.1.6	Draw the Deployment Diagram	3.1.4 FS	Trương Hiển Long	11/15/26	11/16/26	2
3.1.7	Write Architecture Decision Records	3.1.5 FS, 3.1.6 FS	Trần Hoàng Minh Khang	11/17/26	11/18/26	2
3.2.1	Draw the Domain Model	2.5.2 FS, 3.1.2 SS	Lê Đoàn Nhật Huy	11/12/26	11/13/26	2
3.2.2	Draw the Class Diagram	3.2.6 FS, 3.2.5 FS, 3.2.1 FS	Trần Hoàng Minh Khang	11/18/26	11/20/26	3
3.2.3	Write Class Specifications	3.2.2 SS+2d	Trương Hiển Long	11/20/26	11/22/26	3
3.2.4	Draw Sequence Diagrams	3.2.2 SS+2d	Huỳnh Trung Ngân	11/20/26	11/22/26	3
3.2.5	Draw State Machine Diagrams	3.2.1 FS	Huỳnh Trung Ngân	11/14/26	11/15/26	2
3.2.6	Document Design Principles and Patterns	3.2.1 FS, 3.1.4 FS	Trương Hiển Long	11/13/26	11/13/26	1
3.3.1	Draw the Data Diagram	3.2.1 SS+1d	Trương Hiển Long	11/13/26	11/14/26	2
3.3.2	Write Data Specifications	3.3.1 SS+1d	Trương Hiển Long	11/14/26	11/16/26	3
3.3.3	Document Normalisation and Indexes	3.3.2 SS+2d	Trương Hiển Long	11/16/26	11/17/26	2
3.4.1	Prepare the Screen List and Screen Flow	3.1.5 FS, 2.4.2 FS	Trần Hoàng Minh Khang	11/16/26	11/17/26	2
3.4.2	Write Screen Specifications	3.4.1 FS, 3.2.2 FS	Trần Hoàng Minh Khang	11/22/26	11/23/26	3
3.4.3	Complete the Accessibility Checklist	3.4.2 FS	Lê Đoàn Nhật Huy	11/24/26	11/24/26	1
3.5.1	Complete the Traceability Update	3.1.7 FS, 3.2.3 FS, 3.2.4 FS, 3.3.3 FS, 3.4.2 FS, 3.4.3 FF	Huỳnh Trung Ngân	11/23/26	11/24/26	2
3.5.2	Document Changes to the Requirements Since PA2	2.5.2 FS	Lê Đoàn Nhật Huy	11/12/26	11/12/26	1



3.5.3	Review design consistency	3.5.1 FS, 3.5.2 FS, 3.4.3 FS	Trần Hoàng Minh Khang	11/29/26	12/25/26	1
	Milestone: PA3 Submission					
4.1.1	Define Objectives and Scope	3.5.3 FS	Lê Đoàn Nhật Huy	12/26/26	12/26/26	1
4.1.2	Define Test Levels and Test Types	4.1.1 SS	Trần Hoàng Minh Khang	12/26/26	12/26/26	1
4.1.3	Select Test Design Techniques	4.1.2 FS	Lê Đoàn Nhật Huy	12/27/26	12/27/26	1
4.1.4	Specify Test Environment and Test Data	4.1.2 FS	Trương Hiển Long	12/27/26	12/28/26	2
4.1.5	Define Entry and Exit Criteria	4.1.4 FF	Lê Đoàn Nhật Huy	12/28/26	12/28/26	1
4.1.6	Define Defect Management	4.1.2 FS	Huỳnh Trung Ngan	12/27/26	12/27/26	1
4.1.7	Document Schedule, Responsibilities and Tools	4.1.3 FS, 4.1.5 SS, 4.1.6 FS	Huỳnh Trung Ngan	12/28/26	12/28/26	1
4.1.8	Document Test Risks	4.1.7 SS	Huỳnh Trung Ngan	12/28/26	12/28/26	1
4.2.1	Define Equivalence Partitions and Boundary Values	4.1.3 FS	Trần Hoàng Minh Khang	12/28/26	12/29/26	2
4.2.2	Create Decision Tables	4.1.3 SS	Trần Hoàng Minh Khang	12/27/26	12/28/26	2
4.3.1	Prepare the List of Test Cases	4.2.1 FF, 4.2.2 FS	Lê Đoàn Nhật Huy	12/29/26	12/29/26	1
4.3.2	Write Test Case Specifications	4.3.1 SS, 4.1.5 FS, 4.1.6 FS	Lê Đoàn Nhật Huy	12/29/26	12/31/26	3
4.3.3	Complete Requirements–Test Traceability	4.3.2 FS, 4.2.8 FS	Huỳnh Trung Ngan	1/2/26	1/2/26	1
	Milestone: PA4 Submission					
5.1.1	Set up the coding environment	4.3.3 FS	Trương Hiển Long	12/3/26	12/3/26	1
5.1.2	Set up the testing environment	5.1.1 FS	Trương Hiển Long	12/4/26	12/4/26	1
5.1.3	Define coding standards	5.1.1 SS, 3.2.7 FS	Huỳnh Trung Ngan	12/3/26	12/3/26	1
5.1.4	Configure the CI pipeline	5.1.2 FS, 5.1.3 FS	Trương Hiển Long	12/7/26	12/8/26	2
5.2.1	Implement navigation and routing	5.1.3 SS, 3.4.2 FS	Trần Hoàng Minh Khang	12/3/26	12/4/26	2
5.2.2	Implement page views and layouts	5.2.1 SS+1d, 3.4.2 FS	Trần Hoàng Minh Khang	12/4/26	12/7/26	4
5.2.3	Implement accessibility requirements	5.2.2 SS+3d	Trần Hoàng Minh Khang	12/7/26	12/8/26	2
5.2.4	Integrate frontend and backend APIs	5.4.2 FS, 5.2.2 FS, 5.4.3 FF, 5.4.4 FF, 5.4.5 FF	Huỳnh Trung Ngan	12/8/26	12/10/26	3
5.3.1	Set up the database	5.1.1 SS	Trương Hiển Long	12/3/26	12/3/26	1
5.3.2	Configure database access	5.3.1 FF	Trương Hiển Long	12/3/26	12/3/26	1
5.3.3	Implement the database schema	5.3.2 FS	Trương Hiển Long	12/4/26	12/5/26	2
5.3.4	Seed test data	5.3.3 FS	Lê Đoàn Nhật Huy	12/6/26	12/6/26	1
5.3.5	Configure file storage	5.3.2 FS, 3.3.2 FS	Trương Hiển Long	12/6/26	12/7/26	2
5.4.1	Implement authentication and authorisation (F01)	5.3.3 SS+1d, 5.3.3 FF, 5.1.3 FS	Trương Hiển Long	12/9/26	12/11/26	3
5.4.2	Implement course management (F02)	5.4.1 SS+1d, 5.4.1 FF	Huỳnh Trung Ngan	12/6/26	12/7/26	2
5.4.3	Implement enrollment and rosters (F03)	5.4.2 SS+1d, 5.4.2 FF, 5.4.2 FF	Lê Đoàn Nhật Huy	12/7/26	12/8/26	2
5.4.4	Implement course materials (F04)	5.4.2 SS+1d, 5.3.5 FF, 5.4.2 FF	Trương Hiển Long	12/7/26	12/9/26	3
5.4.5	Implement assignments and submissions (F05)	5.4.2 SS, 5.4.1 FF, 5.4.2 FF, 5.3.5 FF	Huỳnh Trung Ngan	12/6/26	12/9/26	4
5.4.6	Implement secondary features (F06–F09, optional)	5.9.2 FS	Huỳnh Trung Ngan	12/22/26	12/22/26	6
5.5.1	Review code and resolve static analysis issues	5.2.4 FS, 5.2.4 FS, 5.5.6 SS+2d, 5.5.6 FF	Huỳnh Trung Ngan	12/22/26	12/23/26	2
5.5.2	Write and execute unit tests	5.4.3 FS, 5.4.4 FS, 5.4.5 FS, 5.2.2 FS	Trương Hiển Long	12/22/26	12/22/26	3
5.5.3	Automate system test scenarios	5.2.4 SS+1d, 5.2.4 FF, 5.2.2 FS	Lê Đoàn Nhật Huy	12/9/26	12/22/26	3
5.5.4	Execute functional integration, system and acceptance tests	5.2.4 FS, 5.4.3 FS, 5.4.4 FS, 5.4.5 FS, 5.3.4 FS, 5.5.2 FF, 5.5.2 FF, 5.5.6 FF	Lê Đoàn Nhật Huy	12/22/26	12/23/26	3
5.5.5	Fix defects and run regression tests	5.5.4 SS+2d, 5.5.4 FF, 5.5.7 FF	Huỳnh Trung Ngan	12/23/26	12/25/26	3
5.5.6	Implement Non-Functional Requirements	5.2.4 FS, 5.2.3 FS	Trương Hiển Long	12/22/26	12/23/26	3
5.5.7	Execute non-functional tests	5.5.6 SS+2d, 5.5.6 FF, 5.5.1 FF	Lê Đoàn Nhật Huy	12/23/26	12/24/26	2
5.5.8	Verify test exit criteria	5.5.5 FF, 5.5.4 FS, 5.5.7 FS	Lê Đoàn Nhật Huy	12/25/26	12/25/26	1
5.6.1	Configure the deployment environment	5.1.4 FS, 5.3.5 FS, 5.3.2 FS	Trương Hiển Long	12/22/26	12/22/26	2
5.6.2	Deploy and verify the system	5.6.6 FS, 5.5.8 FS, 5.6.1 FS	Trương Hiển Long	12/26/26	12/26/26	1
5.6.3	Prepare the Release and User Guide	5.6.6 SS, 5.6.2 FF, 5.6.5 FF	Trương Hiển Long	12/25/26	12/26/26	2
5.6.4	Record and publish the Working Software Video	5.6.2 FF, 5.6.5 FF, 5.2.3 FS	Trần Hoàng Minh Khang	12/25/26	12/26/26	2
5.6.5	Prepare access details and demo accounts	5.6.2 FF	Lê Đoàn Nhật Huy	12/26/26	12/26/26	1
5.6.6	Create the release tag	5.5.8 FF, 5.5.1 FS, 5.5.7 FS	Trương Hiển Long	12/25/26	12/25/26	1
5.7.1	Document Source Code and Structure	5.5.6 FS, 5.5.1 FS	Trương Hiển Long	12/24/26	12/24/26	1
5.7.2	Update Feature Status	5.5.8 FF	Huỳnh Trung Ngan	12/25/26	12/25/26	1
5.7.3	Document Coding Standards and Code Review	5.5.1 FS	Huỳnh Trung Ngan	12/24/26	12/24/26	1
5.7.4	Document Static Analysis	5.5.1 FS	Huỳnh Trung Ngan	12/24/26	12/24/26	1
5.7.5	Document Deviations from the Design	5.5.5 FF	Trần Hoàng Minh Khang	12/25/26	12/25/26	1
5.8.1	Write the Test Execution Summary	5.5.8 FS, 5.8.2 FS, 5.8.3 FS, 5.8.4 FS, 5.5.7 FS	Lê Đoàn Nhật Huy	12/26/26	12/26/26	1
5.8.2	Compile Test Case Results and the Defect List	5.5.5 FF	Lê Đoàn Nhật Huy	12/25/26	12/25/26	1
5.8.3	Document Unit Tests and Coverage	5.5.2 FS	Trương Hiển Long	12/23/26	12/23/26	1
5.8.4	Document Automated System Tests	5.5.4 FS, 5.5.3 FS	Lê Đoàn Nhật Huy	12/24/26	12/24/26	1
5.8.5	Document Continuous Integration	5.1.4 FS, 5.2.4 FF	Trương Hiển Long	12/22/26	12/22/26	1
5.8.6	Document the Deployment Environment	5.6.2 FF, 5.6.5 FF	Trương Hiển Long	12/26/26	12/26/26	1
5.9.1	Document Plan vs. Actual	5.6.2 FF, 5.8.1 FF, 5.6.3 FF, 5.6.4 FF	Huỳnh Trung Ngan	12/26/26	12/26/26	1
5.9.2	Document Process Reflection and Improvements	5.9.1 FF, 5.8.6 FF	Huỳnh Trung Ngan	12/26/26	12/26/26	1
5.9.3	Document AI Usage Lessons	5.5.4 FS, 5.5.7 FS	Lê Đoàn Nhật Huy	12/25/26	12/25/26	1



Milestone: PAS Submission						
6.1.1	Prepare presentation slides	5.9.2 FS, 5.6.3 FS, 5.6.4 FS	Everyone	12/23/26	12/29/26	3
6.1.2	Prepare the live demo	5.6.2 FS, 5.6.3 FS	Truong Hiến Long	12/23/26	12/23/26	1
6.1.3	Rehearse the presentation and Q&A	6.1.1 FS, 6.1.2 FS, 5.4.6 FF	Everyone	12/22/26	12/22/26	1
6.1.4	Deliver the presentation and demo	6.1.3 FS, 5.4.6 FS	Everyone	12/23/26	12/23/26	1
Milestone: End of project						